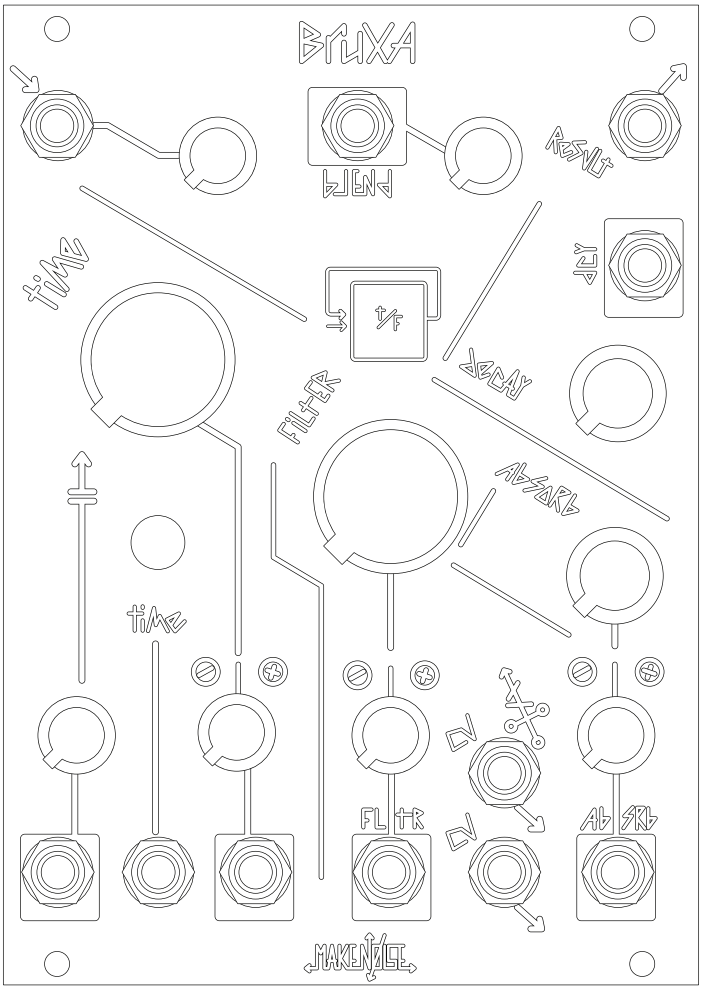
BRUXA



MAKENOISE

TABLE OF CONTENTS

1. LIMITED WARRANTY
2. INSTALLATION
3. INTRODUCTION
4. PANEL CONTROLS
5. HISTORY
6. PATCH IDEAS

LIMITED WARRANTY

Make Noise warrants this product to be free of defects in materials or construction for a period of one year from the date of purchase (proof of purchase/invoice required).

Malfunction resulting from wrong power supply voltages, backwards or reversed eurorack bus board cable connection, abuse of the product, removing knobs, changing faceplates, or any other causes determined by Make Noise to be the fault of the user are not covered by this warranty, and normal service rates will apply.

During the warranty period, any defective products will be repaired or replaced, at the option of Make Noise, on a return-to-Make Noise basis with the customer paying the transit cost to Make Noise.

Make Noise implies and accepts no responsibility for harm to person or apparatus caused through operation of this product.

Please contact technical@makenoisemusic.com with any questions, Return To Manufacturer Authorization, or any needs & comments.

http://www.makenoisemusic.com



About This Manual:

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Illustration and layout by Lewis Dahm

INSTALLATION

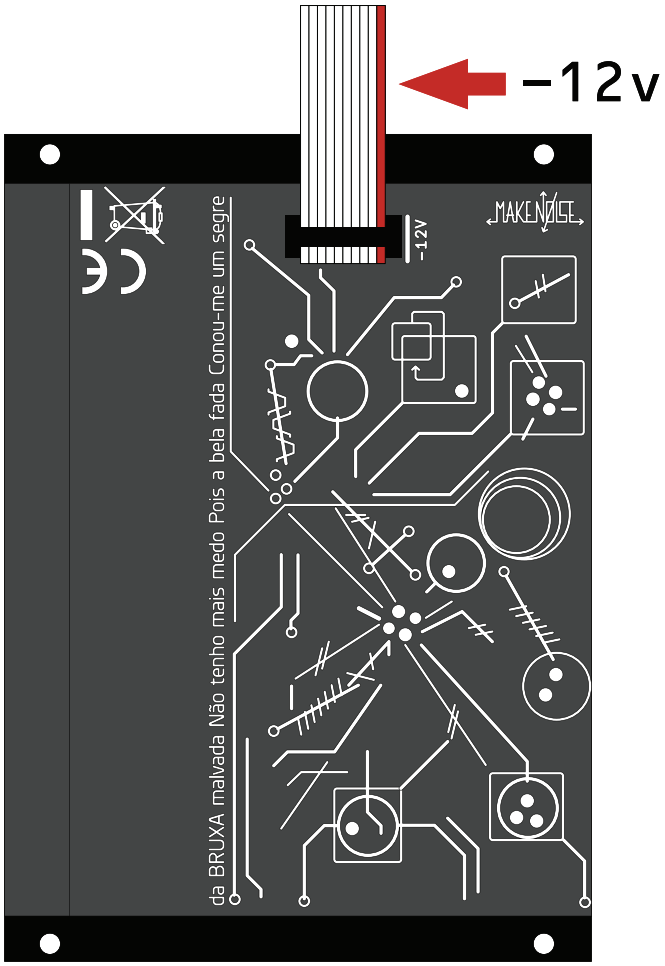
Electrocution hazard!

Always turn the Eurorack case off and unplug the power cord before plugging or unplugging any Eurorack bus board connection cable. Do not touch any electrical terminals when attaching any Eurorack bus board cable.

The Make Noise Bruxa is an electronic music module requiring 140mA of +12VDC and 90mA of -12VDC regulated voltage and a properly formatted distribution receptacle to operate. It must be properly installed into a Eurorack format modular synthesizer system case.

Go to http://www.makenoisemusic.com/ for examples of Eurorack Systems and Cases.

To install, find 18HP in your Eurorack synthesizer case, confirm proper installation of Eurorack bus board connector cable on backside of module (see picture below), plug the bus board connector cable into the Eurorack style bus board, minding the polarity so that the RED stripe on the cable is oriented to the NEGATIVE 12 Volt line on both the module and the bus board. On the Make Noise 6U or 3U Busboard, the negative 12 Volt line is indicated by the white stripe.



Please refer to your case manufacturer's specification for location of the negative supply.

INTRODUCTION

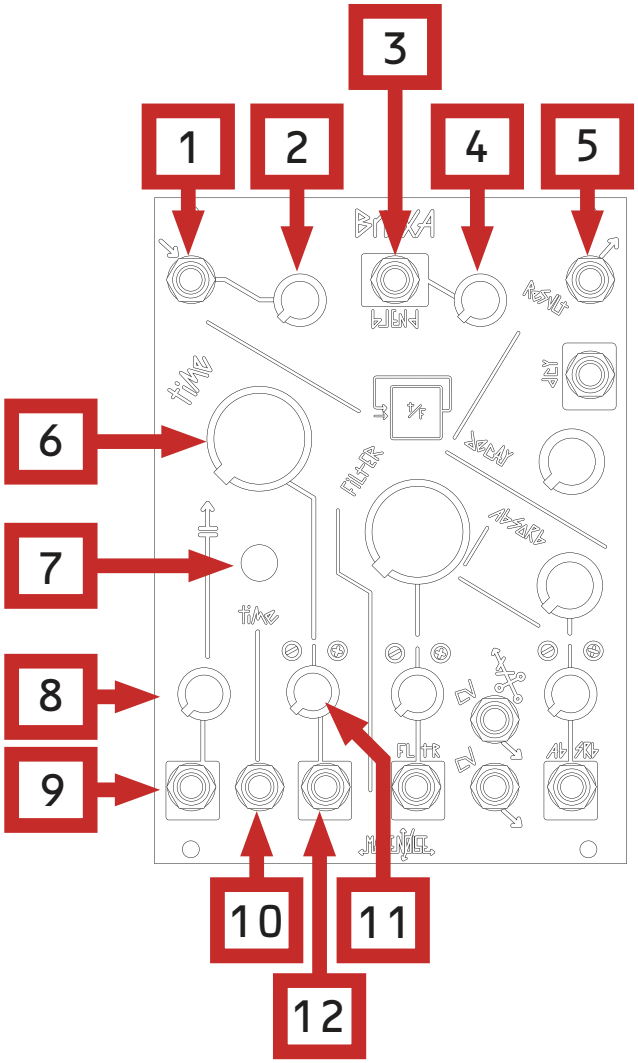
The **Bruxa** music synthesizer module is an audio alchemical experiment. Spill your sound into time, and let it decay through the filters to hear the results.

Bruxa is a collaboration between Make Noise and Alessandro Cortini. Inspired by Cortini's music and sonic experimentations, Bruxa embodies the seen and unseen, the sign and magic, the alchemy of sound.

Bruxa is a new evolution of Tony and Alessandro's original prototype circuits which eventually became the **Strega's** Time/Filter Experiment. It is a multi-tap delay line with over a dozen unique feedback paths and multiple filters within those feedback paths aligned to allow for accumulation of noise, saturation and signal degradation.

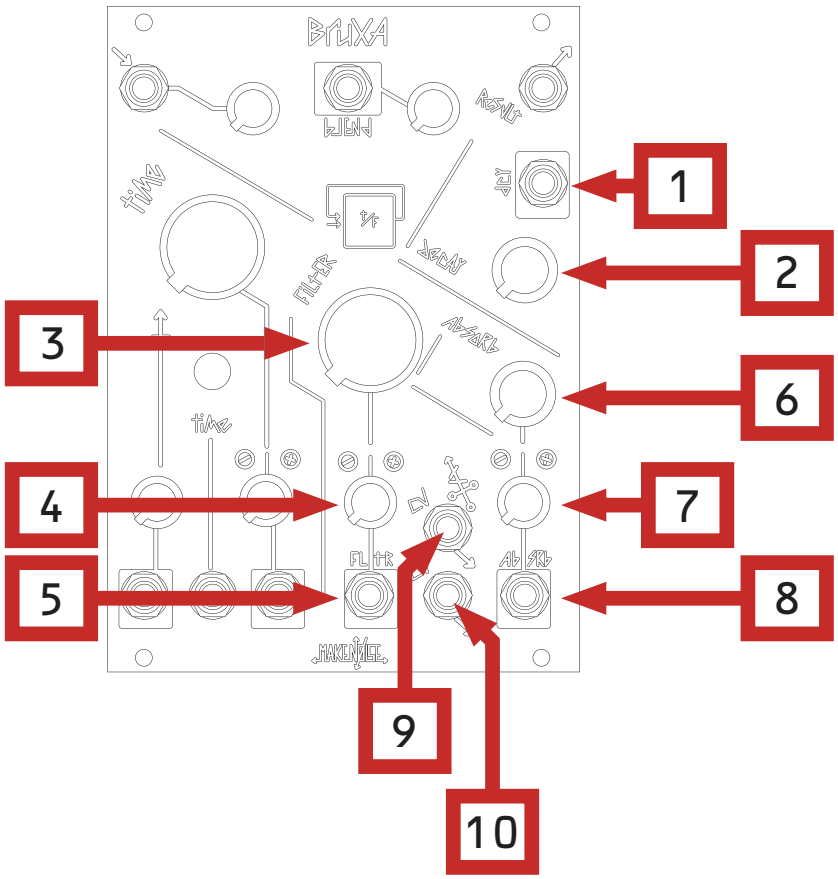
While Strega is a complete synthesizer in a tabletop form factor, Bruxa is a Eurorack module designed to be used as one part of a Eurorack modular synthesizer. You will discover a number of differences in layout, signals, and sound. Living inside the Eurorack modular system, Bruxa encourages you to patch your own sound and modulation sources, yielding results surprisingly distinct from Strega.

PANEL CONTROLS



1. **Signal Input:** Signal Input
2. **Input Level Panel Control:** Sets the level of the incoming signal at the Signal Input.
3. **Blend CV Input:** Control voltage input for Blend.
4. **Blend Combo Pot:** Blends between the Signal Input and the Time/Filter experiment (aka Dry/Wet Mix).
5. **Output:** Audio output. Modular Level.
6. **Time Coarse:** Wide range adjustment for the rate at which audio signals are poured into the filter.
7. **Time Fine:** Short range precision adjustment for the rate at which audio signals are poured into the filter.
8. **Time Modulation attenuator:** Sets the depth of Time Modulation.
9. **Time Modulation CV Input:** Attenuated AC coupled modulation input for Time. Typical input 10vpp.
10. **Time Unity CV In:** Full Scale control input for Time, range +/-10vDC. Be careful.
11. **Time CV Attenuverter:** Bipolar attenuator for Time CV In.
12. **Time CV In:** Attenuated direct coupled, bipolar CV input for Time.

PANEL CONTROLS



1. **Decay CV In:** Direct coupled CV input for Decay.
2. **Decay Panel Control:** Sets the iterations of Time/Filter experiment. Ranges from a handful to infinity.
3. **Filter:** Removes unwanted substance from the results of Time/Filter experiment.
4. **Filter CV Attenuverter:** Bipolar attenuator for Filter CV In.
5. **Filter CV In:** Attenuated direct coupled bipolar for Filter.
6. **Absorb Panel Control:** Absorbs Filtered substances into the earth.
7. **Absorb CV Attenuverter:** Attenuated direct coupled bipolar for Absorb.
8. **Absorb CV Input:** Direct coupled CV Input for Absorb.
9. **CV Out:** DC Voltage Feedback signal from the Time/ Filter circuit. Indicated visually on the T/F Activity Window.
10. **Interference CV:** Control signal derived from the Time/ Filter experiment thus creating an additional feedback path when patched into Time, Filter or Absorb, allowing the module to influence itself. Indicated visually on the T/F Activity Window.

HISTORY

Elements of Bruxa will be familiar if you have used Strega, but the two instruments have each taken a unique development path from the prototype circuit that is their common ancestor. Strega's Time/Filter experiment became thoroughly merged with and tuned to the circuits that would become the Strega Tones and its integration with line level instruments, whereas Bruxa's Time/Filter experiment is designed to intermingle more smoothly with a wide range of sonic material from within a modular synthesizer.

Accordingly, there are differences in Bruxa's control locations and ranges, the derivation of its CV outputs, its musical response to synthesists' gestures, and its overall sonic character. Not all these differences are known, but with time they will be intuited and perhaps understood.

THE BRUXA EXPERIMENT

The most well documented Bruxa experiment begins with the input. Point all controls to midnight, and send a sound of your choice into the input. Adjust the input level to taste. Hear how changes in your source sound affect the Time/Filter experiment. Low or quiet sounds vs. higher or louder pitches. Turn up Blend, and sweep the Time control through its entire range. At around 9:00, you should hear an avalanche of noise. Turn up the Absorb control and sweep the filter. Patch the CV outputs to the Filter or Absorb inputs or wherever you like.

PATCH IDEAS

Wow & Flutter

```
Sound Source -> Signal Input.
CV Out -> Time CV In.
```

Set Time panel control and CV attenuverter to 12:00, then adjust slightly to taste. Slower times are more “tape-like” but also potentially noisier.

Noise

```
Mult Output to Monitor, and also to Signal Input.
Interference CV Out -> Absorb.
```

Set Mix fully wet. Adjust all controls to taste.

Wind & Wave

```
Interference CV Out -> Signal Input.
CV Out -> Filter CV Input.
```

Set Mix and Input Level fully clockwise, and turn Time all the way down to begin.